

Practica 4

Sección 3: Pricing Calculator

Escenario	Horas de cómputo	Coste cómputo	Coste disco (72 h)	Coste IP (72 h)	Total estimado
A. VM encendida todo el laboratorio	72	5.86\$ al mes	0	0.26\$ al mes	6.12\$ al mes
B. VM con apagado automático nocturno (≈ 12 h/día)	36	4'13\$ al mes	0	0.13\$ al mes	4.26\$ al mes
C. VM encendida solo 8 h en total	8	2.78\$ al mes	0	0.03\$ al mes	2.81\$ al mes

El apartado configurable de disco está dentro del apartado configurable de la propia VM, por lo que no se indica ningún coste adicional.

Sección 4: Crear Resource Group Etiquetado

Crear Resource Group

Create a resource group ...

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ

Azure for Students

Resource group name * ⓘ

rg-practica04-vm-alumno09

Region * ⓘ

(Europe) West Europe

Añadir Tags

Create a resource group ...

Basics **Tags** Review + create

Apply tags to your Azure resources to logically organize them by categories. A tag consists of a key (name) and a value. Tag names are case-insensitive and tag values are case-sensitive. [Learn more](#)

Name		Value	Resource	
Project	:	Practica04	Resource group	
Department	:	Data	Resource group	
Environment	:	Lab	Resource group	
Owner	:	Alumno09	Resource group	
CostCenter	:	DataNova-Analytics	Resource group	
	:		Resource group	

Review

Create a resource group ...

- Basics
- Tags
- Review + create

 Automation Link

Basics

Subscription	Azure for Students
Resource group name	rg-practica04-vm-alumno09
Region	West Europe

Tags

Project	Practica04
Department	Data
Environment	Lab
Owner	Alumno09
CostCenter	DataNova-Analytics

Sección 5: Crear un Budget específico para el laboratorio

Crear Budget desde el Resource Group

Home > rg-practica04-vm-alumno09

 **rg-practica04-vm-alumno09 | Budgets**   ... 

Resource group

 Search

 Add  Refresh  Feedback  Help 

 Overview

 Activity log

 Access control (IAM)

 Tags

 Resource visualizer

 Events

> Settings

✓ Cost Management

 Cost analysis

 Cost alerts (preview)

 **Budgets** 

 Advisor recommendations

> Monitoring

> Automation

> Help

Scope :  **rg-practica04-vm-alumno09**

 S

Name	↑↓	Scope	↑↓	Reset p
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You do not have any budgets.

Definir Estimación

Hay que introducir el redondeo hacia arriba del valor por tener la VM iniciada 36h.
Por lo que será de 5\$.

Configurar Budget

Home > rg-practica04-vm-alumno09 | Budgets

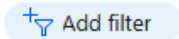
Create budget ...

Budget

Budget scoping

The budget you create will be assigned to the selected scope. Use additional filters like resource groups to have your budget monitor with more granularity as needed.

Scope  rg-practica04-vm-alumno09

Filters 

Budget Details

Give your budget a unique name. Select the time window it analyzes during each evaluation period, its expiration date and the amount.

*Name	Practica4 		
*Reset period ⓘ	Monthly 		
*Creation date ⓘ	2026 	September 	1
*Expiration date ⓘ	2028 	August 	1 

Budget Amount

Give your budget amount threshold

Amount *	5 
----------	--

Configurar las alertas

Home > Resource Manager | Resource groups > rg-practica04-vm-alumno09 | Budgets

Create budget ...

Budget

✓ Create a budget ✓ Set alerts

Configure alert conditions and send email notifications based on your spend.

* Alert conditions

Type	% of budget	Amount	Action group
Actual cost	50	2,5	None
Actual cost	80	4	None
Actual cost	100	5	None
Forecasted cost ▾	100 ✓	5	None ▾
Select type ▾	Enter %	-	None ▾

[Manage action group](#) ⓘ

* Alert recipients (email)

Alert recipients (email)

ikeraimar.gomez-espinoza@tajamar365.com ✓



example@email.com

It is recommended to add `azure-noreply@microsoft.com` to your email allow list to ensure alert mails do not go to your spam folder.

Sección 6: Desplegar la máquina virtual

Crear la VM

Create a virtual machine ...

- Basics
- Disks
- Networking
- Management
- Monitoring
- Advanced
- Tags
- Review + create

 This is a preview of a new Create-VM experience. Features are limited in this experience and all feedback is appreciated. [Click here to access the previous experience.](#)

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#) 

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * 

Azure for Students

Resource group * 

rg-practica04-vm-alumno09

[Create new](#)

Instance details

Virtual machine name * 

vm-practica04-alumno09

Region * 

(Europe) Spain Central

[Deploy to an Azure Extended Zone](#)

Availability options 

No infrastructure redundancy required

Security type 

Standard

Image * 

 Ubuntu Server 24.04 LTS - x64 Gen2

[See all images](#) | [Configure VM generation](#)

Create a virtual machine ...

Basics Disks Networking Management Monitoring Advanced Tags Review + create

VM architecture * ⓘ

- ☐ Arm64
- ☒ x64

Size * ⓘ

Standard_B2s_v2 - 2 vcpus, 8 GiB memory (66,58 US\$/month) ▾

[See all sizes](#) | [Customize cores](#)

Run with Azure Spot discount ⓘ

☐

Administrator account

Authentication type * ⓘ

- ☒ SSH public key
- ☐ Password

ⓘ Azure now automatically generates an SSH key pair for you and allows you to store it for future use. It is a fast, simple, and secure way to connect to your virtual machine.

Username * ⓘ

azureuser

SSH public key source * ⓘ

Generate new key pair ▾

SSH Key Type * ⓘ

- ☒ RSA SSH Format
- ☐ Ed25519 SSH Format

Key pair name * ⓘ

vm-practica04-alumno09_key

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

- ☒ None
- ☐ Allow selected ports

Select inbound ports

Select one or more ports ▾

ⓘ All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

Previous

Next

Review + create

Configuración de Disk

Create a virtual machine ...

- Basics
- Disks**
- Networking
- Management
- Monitoring
- Advanced
- Tags
- Review + create

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Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

 There is a charge for the underlying storage resources consumed by your virtual machine. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host 

☐

 Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size 

Image default (30 GiB)

▼

OS disk type * 

Standard SSD

▼

 The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM 

☒

Enable Ultra Disk compatibility 

☐

 Ultra disk is only supported in Availability Zone(s) 1,3 for the selected VM size Standard_B2s_v2.

Create a virtual machine ...

- Basics
- Disks
- Networking
- Management
- Monitoring
- Advanced
- Tags
- Review + create

Delete with VM

Enable Ultra Disk compatibility

☐

Ultra disk is only supported in Availability Zone(s) 1,3 for the selected VM size Standard_B2s_v2.

Key management

Platform-managed key

▼

Data disks

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching	Delete with VM
-----	------	------------	-----------	--------------	----------------

Create and attach a new disk

Attach an existing disk

Advanced

Ephemeral OS disk

- ☒ None
- ☐ OS cache placement
- ☐ Temp disk placement
- ☐ NVMe placement

Ephemeral OS disks are currently not supported for the selected instance size.

Configuración Networking

Create a virtual machine ...

- Basics
- Disks
- Networking
- Management
- Monitoring
- Advanced
- Tags
- Review + create

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Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more](#) 

Virtual network

Azure Virtual Network (VNet) enables many types of Azure resources to securely communicate with each other, the internet, and on-premises networks. [Learn more about Vnets](#) 

Virtual network *

(new) vnet-spaincentral-1 (rg-practica04-vm-alumno09) 

[Edit virtual network](#)

Subnet *

(new) snet-spaincentral-1 - 172.16.0.0/24 

[Edit subnet](#)

172.16.0.0/24

Network interface

A network interface enables an Azure virtual machine to communicate with internet, Azure, and on-premises resources. A VM can have one or more network interfaces.

Public IP 

(new) vm-practica04-alumno09-ip 

[Create new](#)

 Public IP addresses have a nominal charge. [Learn more](#) 

NIC network security group 

☐ None

-

Create a virtual machine ...

Basics

Disks

Networking

Management

Monitoring

Advanced

Tags

Review + create

NIC network security group ⓘ

- ☐ None
- ☒ Basic
- ☐ Advanced

Public inbound ports * ⓘ

- ☒ None
- ☐ Allow selected ports

Select inbound ports

Select one or more ports

ⓘ All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

Delete public IP and NIC when VM is deleted ⓘ



Enable accelerated networking ⓘ



ⓘ This size requires accelerated networking

Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. [Learn more](#) ↗

Load balancing options ⓘ

- ☒ None
- ☐ Azure load balancer
Supports all TCP/UDP network traffic, port-forwarding, and outbound flows

Configuración Managment

Home > Compute infrastructure | Virtual machines

Create a virtual machine ...

- Basics
- Disks
- Networking
- Management
- Monitoring
- Advanced
- Tags
- Review + create

i This is a preview of a new Create-VM experience. Features are limited in this experience and all feedback is appreciated. [Click here to access the previous experience.](#)

Configure management options for your VM.

Identity

Enable system assigned managed identity **i** ☐

Microsoft Entra ID

Login with Microsoft Entra ID **i** ☐

i RBAC role assignment of Virtual Machine Administrator Login or Virtual Machine User Login is required when using Microsoft Entra ID login. [Learn more](#) **i**

i Microsoft Entra ID login now uses SSH certificate-based authentication. You will need to use an SSH client that supports OpenSSH certificates. You can use Azure CLI or Cloud Shell from the Azure Portal. [Learn more](#) **i**

Auto-shutdown

Enable auto-shutdown **i** ☐

i The selected region does not support auto-shutdown.

Guest OS updates

Enable periodic assessment **i** ☐

Patch orchestration options **i**

Image default **v**

i Some patch orchestration options are not available for this image. [Learn more](#) **i**

Configuración Tags

Home > Compute infrastructure | Virtual machines

Create a virtual machine ...

- Basics
- Disks
- Networking
- Management
- Monitoring
- Advanced
- Tags**
- Review + create

This is a preview of a new Create-VM experience. Features are limited in this experience and all feedback is appreciated. [Click here to access the previous experience.](#)

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name		Value		Resource	
Project	:	Practica04	:	All resources selected	
CostCenter	:	DataNova-Analytics	:	All resources selected	
Department	:	Data	:	All resources selected	
Environment	:	Lab	:	All resources selected	
Owner	:	Alumno09	:	All resources selected	
	:		:	All resources selected	

Review

Home > Compute infrastructure | Virtual machines

Create a virtual machine ...

- Basics
- Disks
- Networking
- Management
- Monitoring
- Advanced
- Tags
- Review + create

[View automation template](#)

Price

1 X Standard B2s_v2
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Subscription credits apply ⓘ
0.0912 USD/hr

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Name

Iker Aimar Gómez-Espinosa Muñoz

Preferred e-mail address

ikeraimar.gomez-espinosa@tajamar365.com

Preferred phone number

+34917571819

Basics

Subscription

Azure for Students

Resource group

rg-practica04-vm-alumno09

Virtual machine name

vm-practica04-alumno09

Region

Spain Central

Availability options

No infrastructure redundancy required

Security type

Standard

Image

Ubuntu Server 24.04 LTS

Size

Standard B2s_v2

VM architecture

x64

Authentication type

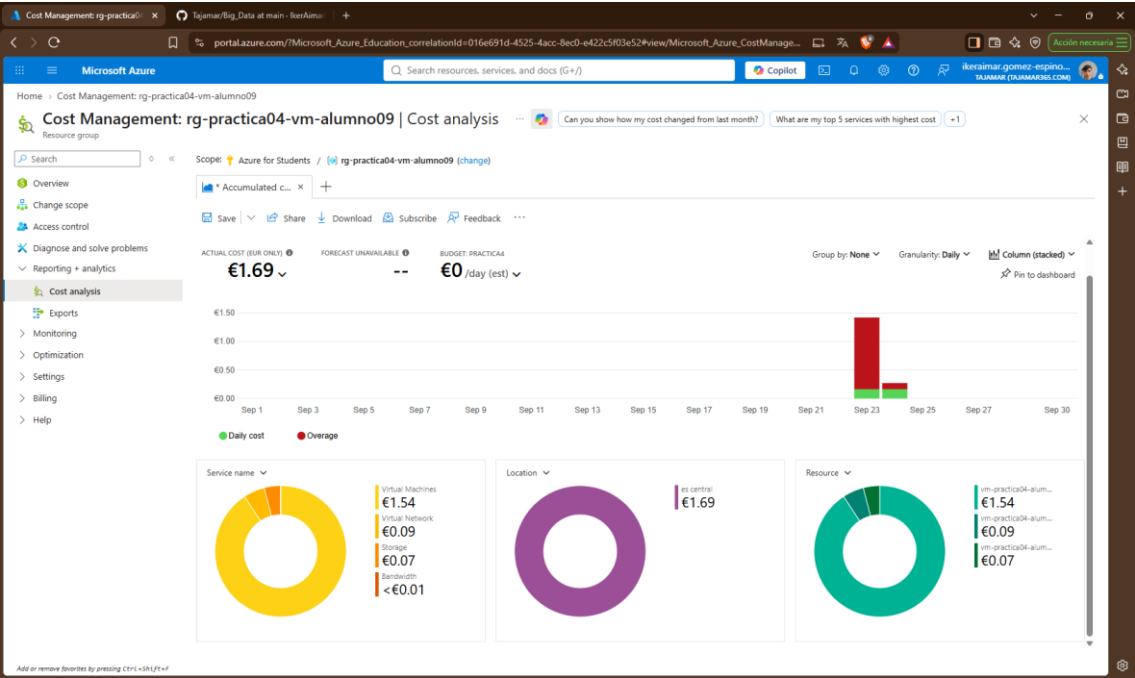
SSH public key

- Previous
- Next
- Create

Recurso	Tipo	¿Tiene los 5 Tags?	¿Genera coste?
vm-practica04-alumno09	Virtual machine	Si	Si
vm-practica04-alumno09	Disk	No	Si
vm-practica04-alumno09	Public IP address	No	Si
vm-practica04-alumno09	Network interface	No	No
vm-practica04-alumno09	Network security group	No	No
vm-practica04-alumno09	Virtual network	No	Si

Sección 8 — Primera revisión en Cost Analysis

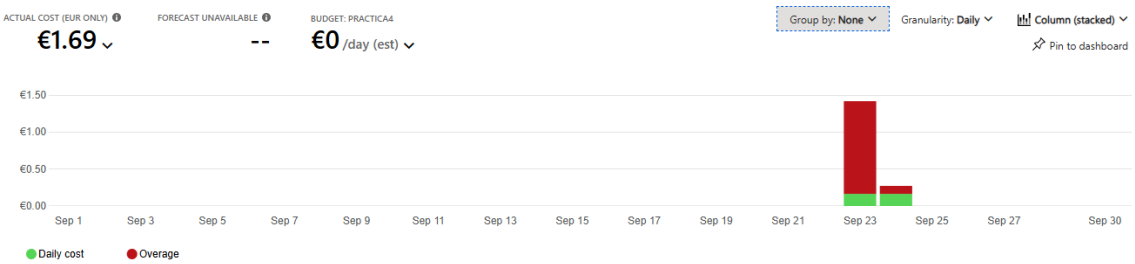
Cost Analysis



Coste Por Recurso

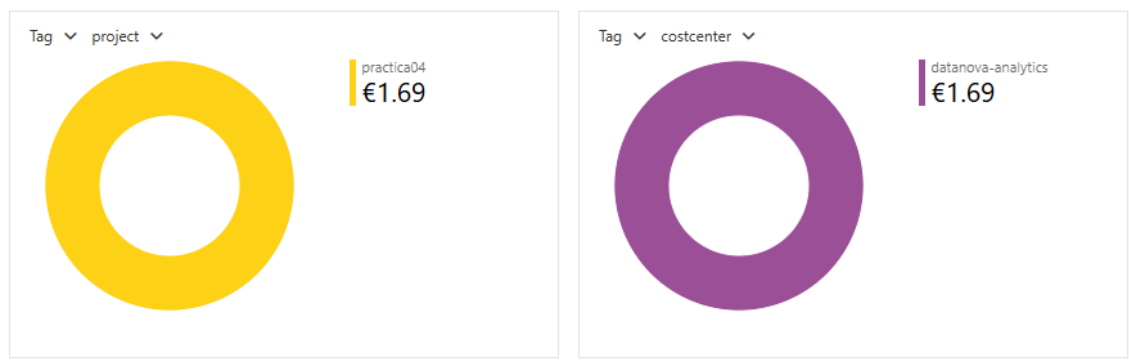
Recurso	Coste acumulado
Máquina virtual	1.54€
Disco	0.07€
IP pública	0.09€
Otros	<0.01€
Total (Actual Cost)	1.69€

Coste Diario En Columna



Las columnas se muestran en formato stacked por que el formato grouped está bloqueado.

Coste Por Tags



Le asigna el precio completo a Project, pero no solo a el, ya que todas las tags están en todos los gastos, pone como si todas las tags gastaran el precio completo.

Consulta de Previsión

Dato	Valor
Actual Cost	1.69€
Forecasted Cost (fin de mes)	--
Budget del laboratorio	5€
¿Actual supera el Budget?	No
¿Forecast supera el Budget?	--

FORECAST UNAVAILABLE ⓘ

BUDGET: PRACTICA4

€0

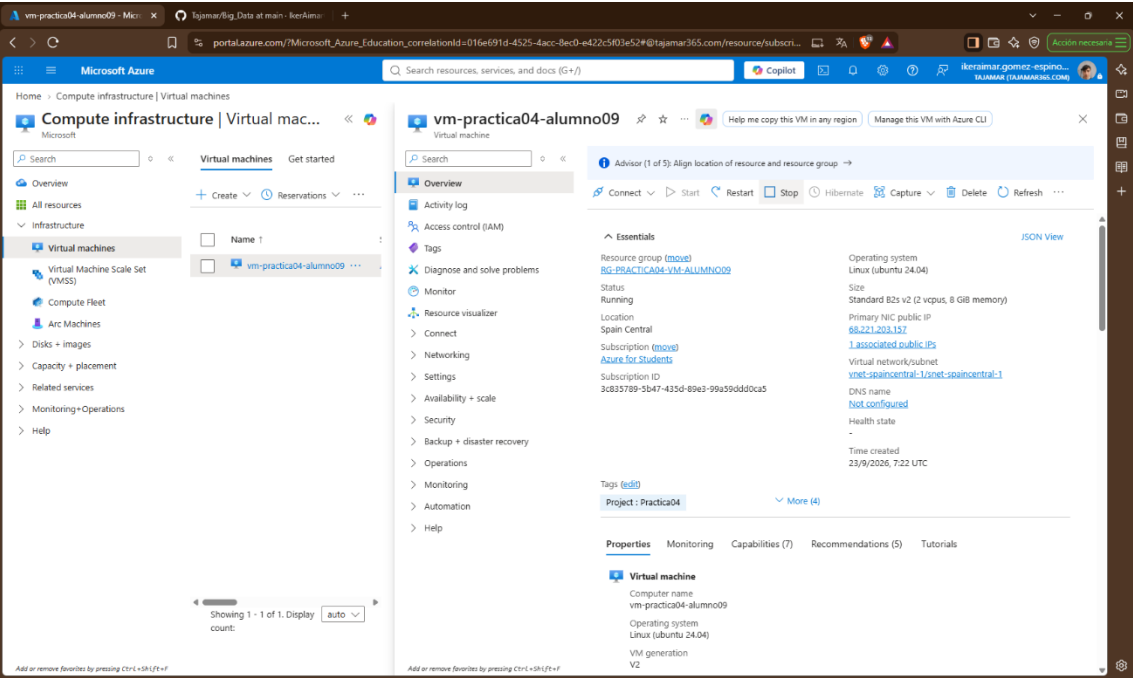
Can't do forecast - 2 points is not enough cost training data. TimeRange: From='9/23/2026 12:00:00 AM', To='9/24/2026 12:00:00 AM' (Code: FailedDependency) [Learn more](#)

Revisar Alertas

Alerta	¿Se ha activado?	Fecha y hora
Actual 50 %	No	24/09/2026 13:26
Actual 80 %	No	24/09/2026 13:26
Actual 100 %	No	24/09/2026 13:26
Forecasted 100 %	No	24/09/2026 13:26

Experimento de Apagado

ApagarMaquina

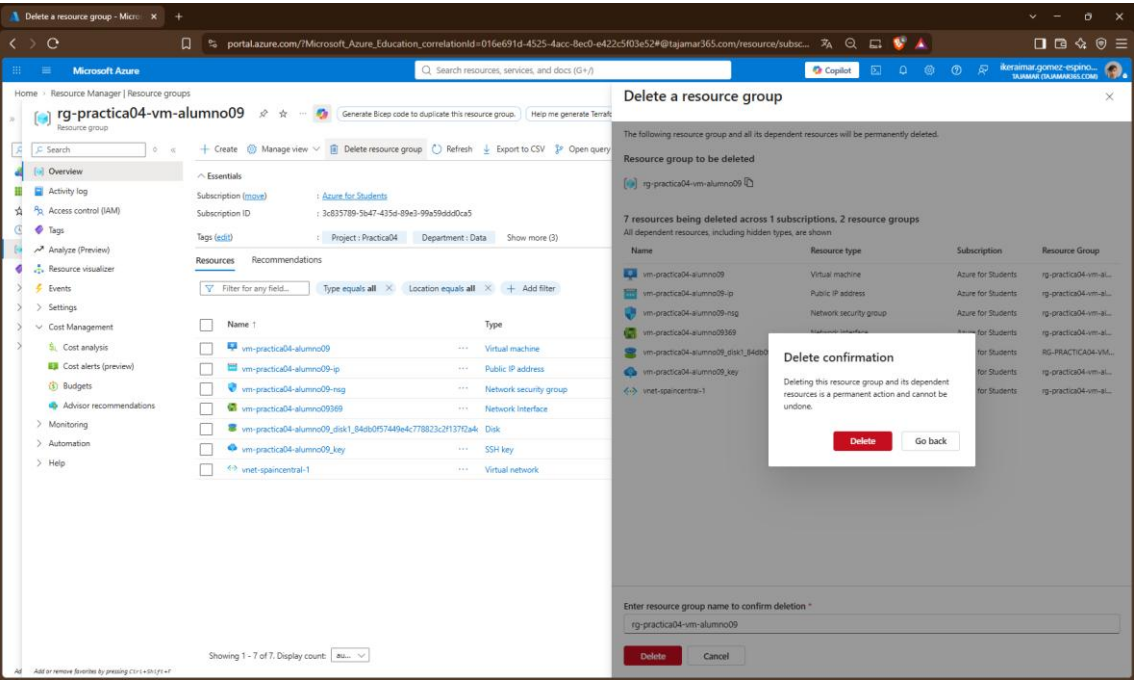


Comparacion estimación y coste

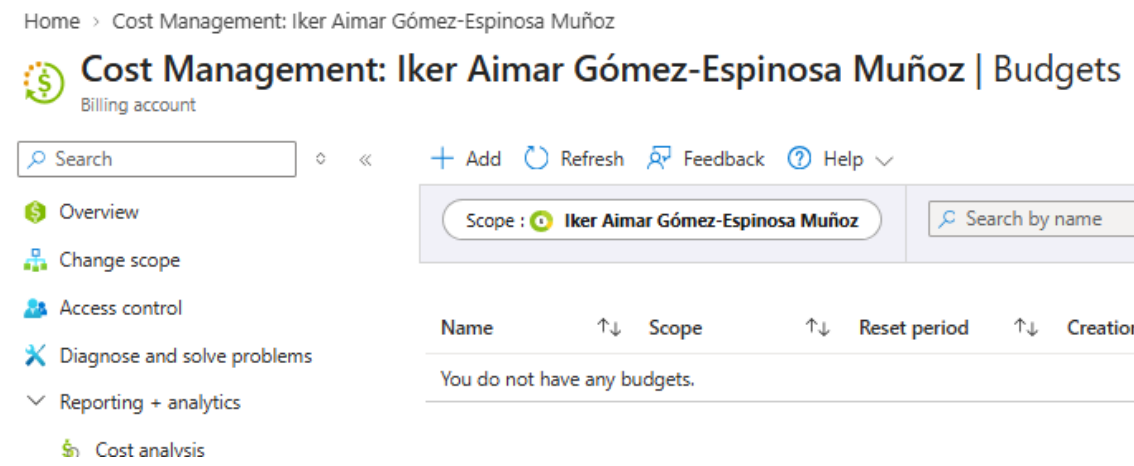
Concepto	Estimado (Sección 3)	Real (Cost Analysis)	Diferencia
Cómputo	0.59	2.26	1.67
Disco	0	0.14	0.14
IP pública	0.02	0.19	0.17
Total	0.62	2.60	1.98

Clean up

Delete Resource Group



No budgets



Informe Final

Elemento	Resultado
Nombre de la práctica	Práctica 04 — VM Linux con control de costes
Resource Group	rg-practica04-vm-alumno09
Servicios utilizados	VM, Discos, IP publicas
Coste estimado antes de desplegar	0.62
Budget disponible	5

Alertas configuradas	Actual 50%, Actual 80%, Actual 100%, Forecasted 100%
Alertas activadas	Forecasted 100%
Coste observado	1.55
Recurso con mayor coste	VM
Recursos eliminados	Resource Group Completo

Ejercicio 13

1. ¿Qué alertas se habrán activado con umbrales Actual 50 %, 80 %, 100 % y Forecasted 100 %?

Unicamente se habría activado la alerta de Forecasted 100%

2. ¿Por qué el Forecasted Cost es tan superior al Actual Cost?

Por que no ha pasado ni un tercio del tiempo sobre el que se proyecta el forecasted cost.

3. ¿Qué parte del gasto de las 3 VMs detenidas se podría eliminar sin borrarlas?

Poniendo las maquinas en Stopped (deallocated) se pararía todo el consumo de la CPU y la RAM sin necesidad de eliminarlas

4. Ordena estas acciones de mayor a menor impacto en el ahorro inmediato:

1. Desasignar las 3 VMs detenidas.
2. Configurar apagado automático en las 5 VMs.
3. Cambiar los discos de Premium SSD a Standard SSD.
4. Eliminar las IP públicas que no se usan.
5. Añadir Tags a todos los recursos.

5. ¿Cuál de las acciones anteriores no ahorra dinero pero es imprescindible para analizarlo?

Añadir Tags a todos los recursos